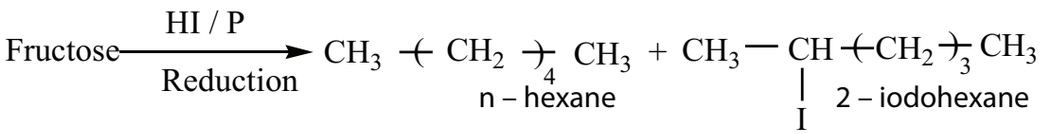
Structure of fructose:

Fructose is the sweetest of all known sugars. It is readily soluble in water. Fresh solution of fructose has a specific rotation -133^0 which changes to – 92^0 at equilibrium due to mutarotation. Similar to glucose the structure of fructose is deduced from the following facts.

1. Elemental analysis and molecular weight determination of fructose show that it has the molecular formula $C_6H_{12}O_6$

2. Fructose on reduction with HI and red phosphorus gives a mixture of n – hexane (major product) and 2 – iodohexane (minor product). This reaction indicates that the six carbon atoms in fructose are in a straight chain.

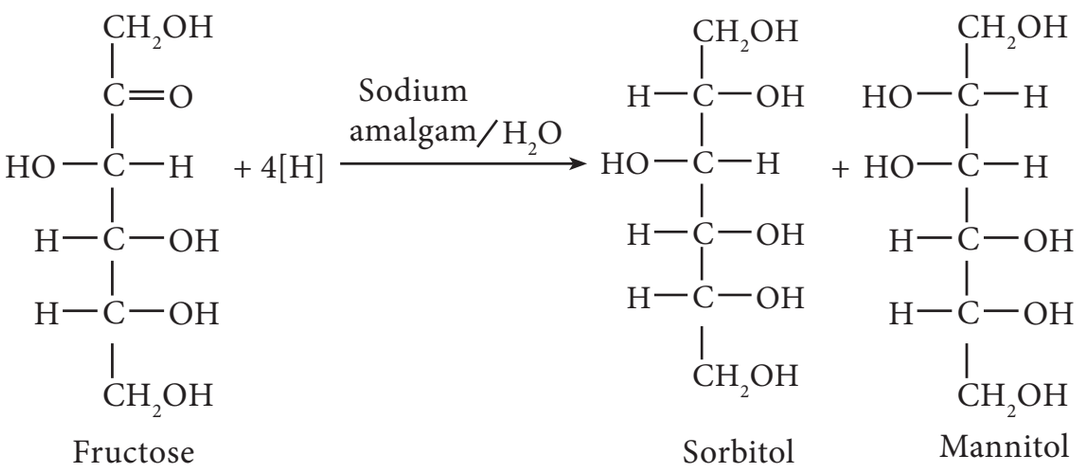


3. Fructose reacts with NH_2OH and HCN. It shows the presence of a carbonyl groups in the fructose.

4. Fructose reacts with acetic anhydride in the presence of pyridine to form penta acetate. This reaction indicates the presence of five hydroxyl groups in a fructose molecule.

5. Fructose is not oxidized by bromine water. This rules out the possibility of absence of an aldehyde (-CHO) group.

6. Partial reduction of fructose with sodium amalgam and water produces mixtures of sorbitol and mannitol which are epimers at the second carbon. New asymmetric carbon is formed at C-2. This confirms the presence of a keto group.



7. On oxidation with nitric acid, it gives glycollic acid and tartaric acids which contain smaller number of carbon atoms than in fructose.

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